

# Coarse-to-Fine-Grained Image Classification:

## A Comparative Study of Feature Representations and Classifiers

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### 1. Introduction

The difficulty of image classification depends strongly on category granularity. Coarse-grained categories can be distinguished by low-level statistics such as color and shape, while fine-grained categories share the overall form and differ only in local details, requiring discriminative features at the semantic level.

This project validates the phenomenon through two contrasting tasks. Task 1 (coarse-grained): 10 visually distinct animal classes, 6 from CIFAR-10 and 4 from public datasets, with 64x64 images and 3,750 training samples. Task 2 (fine-grained): 10 bird species from CUB-200-2011, 128x128 images, only 417 training samples, with some pairs (e.g., two seagulls) nearly indistinguishable.

We applied a unified process on both tasks: evaluating Logistic Regression, SVM with RBF kernel, and Random Forest under 5-fold stratified cross-validation, in two feature spaces (provided manual features and the 2,048-dimensional embedding from an ImageNet-pretrained ResNet50 [Deng et al., 2009; He et al., 2016]). We further engineered hand-crafted HSV color features, performed GridSearchCV on Task 2's best model, and conducted error analysis based on out-of-fold predictions.

Main results: manual feature baselines reached 54.7% on Task 1 and 34.1% on Task 2. ResNet50 embeddings raised these to 91.5% and 86.3%, with a larger absolute

gain on Task 2 (+52 pp) than on Task 1 (+37 pp). Seven of the eight most-confused class pairs on Task 2 match the visual predictions we made from sample images before modelling.

The report is organised as follows: Methods (Section 2), Results (Section 3), Discussion (Section 4), Conclusion (Section 5), and References (Section 6).

### 2. Methodology

#### 2.1 Experimental Design

To separate the contributions of feature quality and classifier choice, we apply a 2 x 3 design: two feature spaces (handcrafted / ResNet50 embedding) x three classifiers (linear / kernel / tree ensemble), giving six (feature, model) combinations per task. This supports three comparisons: (i) fixing the classifier and varying features measures the relative contribution of each feature space; (ii) fixing features and varying classifiers identifies the best classifier for that space; (iii) fixing the (feature, model) combination and comparing across tasks quantifies the inherent difficulty difference between Task 2 and Task 1.

#### 2.2 Data and Preprocessing

Both tasks share the same layout: training/testing images, metadata CSVs, and three precomputed feature sets, namely color histogram (96-d), HOG-PCA (100-d), and additional statistics (23-d, mainly edge density, texture variance, mean channel values).

Color histograms lie in  $[0, 1]$  and HOG-PCA is PCA-centered, but the additional statistics span  $[0, 28785]$  on Task 1. For scale-sensitive models (LR, SVM), unscaled features cause this block to dominate the decision boundary. We therefore wrap `StandardScaler` with each classifier in `sklearn.Pipeline` [Pedregosa et al., 2011], so the scaler fits only on each CV training fold, preventing validation-fold leakage. Random Forest splits on thresholds and is scale-invariant, so no scaler is added.

### 2.3 Feature Representations

(A) Manual feature (baseline): the three provided feature groups are concatenated into a 219-dimensional vector.

(B) ResNet50 CNN embedding (transfer learning): we use `torchvision`'s ResNet50 [He et al., 2016] with ImageNet-1K v2 weights [Deng et al., 2009], replace the final classification layer `fc` with `nn.Identity()`, and take the 2,048-dimensional output after average pooling. ResNet50 is used purely as a frozen feature extractor. All images are preprocessed with the standard ImageNet pipeline (resize to 224x224 and normalise with ImageNet mean and std) before being fed to the network.

(C) Hand-crafted HSV color features: to satisfy the project's requirement of engineering additional feature representations beyond the provided ones, we extract a 14-dimensional HSV descriptor for each image: mean and standard deviation of the H, S, V channels (6 features) plus an 8-bin histogram of the hue channel (8 features). HSV separates color information (H, S) from intensity (V), which is more discriminative than raw RGB for many color-distinct classes.

We also test a hybrid configuration, CNN + traditional (2,267 dimensions), to check whether the manual features add complementary signal beyond the CNN.

### 2.4 Classifiers

We select three classifiers with distinct inductive biases.

- Logistic Regression (L2 regularisation, multinomial,  $C=1.0$ ): the simplest linear model. Its performance directly reflects the discriminative power of the feature space.
- SVM with RBF kernel ( $C=10$ , `gamma='scale'`) [Cortes & Vapnik, 1995]: introduces a nonlinear decision boundary. Its gain over LR shows whether non-linearity adds value in that feature space.
- Random Forest (500 trees) [Breiman, 2001]: a bagging ensemble that reduces variance of single trees and is robust under small samples (e.g., Task 2's 417 images).

Together they form a complexity progression (linear, kernel, ensemble), letting us diagnose whether the performance bottleneck lies in the features or in the model.

### 2.5 Evaluation Protocol

All cross-validations use 5-fold StratifiedKFold (`shuffle=True`, `random_state=42`) to keep each fold's class distribution consistent with the training set. We report two complementary metrics: Accuracy, which matches the Kaggle score; and Macro-averaged F1, which averages per-class F1 equally, preventing dominant classes from masking small-class performance (e.g., Cardinal has only 39 training samples in Task 2).

For the error analysis in Section 4, we use `cross_val_predict` to obtain out-of-fold (OOF) predictions in a single pass. From these we directly compute confusion matrices, per-class precision/recall/F1, and misclassified-sample indices, without re-running CV.

## 2.6 Hyperparameter Tuning

After the baseline, we ran `GridSearchCV` on Task 2's best Random Forest (CNN + traditional): 5-fold inner CV with macro-F1 scoring, grid `n_estimators` in {200, 500}, `max_depth` in {None, 20}, `min_samples_leaf` in {1, 3}. Only Task 2 was tuned: Task 1's best model (LR) already reached 91.5% with default `C=1.0`, and Task 2's small training set is more likely to benefit from regularisation.

## 2.7 Final Submission

After selecting the best model per task, we re-fit it on the full training set (CV uses 4/5 per fold), predict the corresponding test set, and output the result as CSV (`image_id`, `class_id`) for Kaggle.

## 3. Results

### 3.1 Manual Feature Baseline

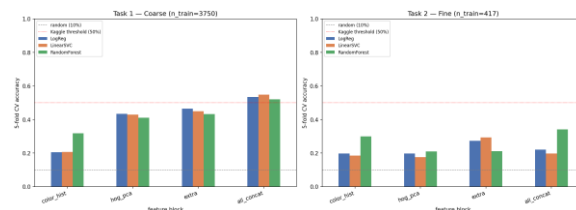
Figure 1 summarises the 5-fold CV accuracy of 24 baseline experiments (3 models x 4 feature combinations x 2 tasks).

Task 1 (coarse-grained): the best baseline is LinearSVC + `all_concat` at 54.7% +/- 2.1%, just crossing the 50% Kaggle threshold. Single-feature ranking is `extra` (46.5%) > `hog_pca` (43.3%) > `color_hist` (31.8%); feature concatenation gives all models a 7-10 pp improvement.

Task 2 (fine-grained): the best baseline is RandomForest + `all_concat` at 34.1% +/- 4.1%, far below 50%. Ranking is

`color_hist` (30.0%) > `extra` (29.2%) > `hog_pca` (20.9%), with HOG nearly losing discriminative power on Task 2. Notably, feature concatenation improves RF by +13 pp but reduces LR and LinearSVC by 5-10 pp, a contrast we discuss in Section 4.

Adding the engineered HSV features (14-d) to the 219-d handcrafted vector gave small but consistent gains on the linear models: Task 1 LinearSVC improved from 54.7% to 55.3% and Task 2 Logistic Regression from 22.1% to 24.7%. Random Forest did not benefit and slightly decreased on Task 2 (34.1% to 31.4%), since its implicit feature selection already captures the most informative color signals from the provided color histogram.



**Figure 1.** Five-fold CV accuracy of three baseline classifiers on Task 1 (left,  $n=3,750$ ) and Task 2 (right,  $n=417$ ) across four feature configurations.

### 3.2 Final Models on CNN Embedding

Table 1 lists the OOF accuracy and macro-F1 for the 12 final experiments (3 models x 2 feature spaces x 2 tasks).

| Tas k | Mode l     | Features       | Accurac y | Macro -F1 |
|-------|------------|----------------|-----------|-----------|
| 1     | LR         | cnn_only       | 0.915     | 0.915     |
| 1     | LR         | cnn_plus_tra d | 0.915     | 0.915     |
| 1     | SVM (RBF ) | cnn_only       | 0.880     | 0.883     |
| 1     | SVM (RBF ) | cnn_plus_tra d | 0.880     | 0.883     |
| 1     | RF         | cnn_only       | 0.887     | 0.887     |
| 1     | RF         | cnn_plus_tra d | 0.894     | 0.893     |

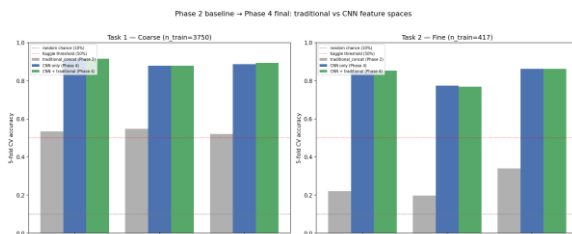
|   |           |               |       |       |
|---|-----------|---------------|-------|-------|
| 2 | LR        | cnn_only      | 0.844 | 0.844 |
| 2 | LR        | cnn_plus_trad | 0.854 | 0.854 |
| 2 | SVM (RBF) | cnn_only      | 0.775 | 0.781 |
| 2 | SVM (RBF) | cnn_plus_trad | 0.770 | 0.775 |
| 2 | RF        | cnn_only      | 0.863 | 0.863 |
| 2 | RF        | cnn_plus_trad | 0.863 | 0.863 |

**Table 1. Five-fold CV results for the 12 final (model, feature) combinations. Best on Task 1: LR + cnn\_only (0.915). Best on Task 2: RF + cnn\_plus\_trad (0.863).**

Task 1: the best result is LR + cnn\_only = 91.5% (macro-F1 = 0.915). The three classifiers perform very similarly (LR > RF 88.7% > SVM 88.0%), indicating that the ResNet50 embedding makes classes nearly linearly separable, so more complex non-linear models extract little additional information.

Task 2: the best result is RF + cnn\_plus\_trad = 86.3% (macro-F1 = 0.863). Model differences are substantial: RF (86.3%) > LR (84.4%) > SVM (77.5%).

Comparing baseline and CNN features: Task 1 improves by +36.8 pp and Task 2 by +52.2 pp. The CNN + traditional concatenation brings only minor additional gain ( $\leq 1$  pp) across all models, indicating that the manual features' information is largely subsumed by the ResNet50 embedding.



**Figure 2. Comparison of baseline (grey) versus CNN-only (blue) and CNN + traditional (green) on Task 1 (left) and Task 2 (right).**

### 3.3 Hyperparameter Tuning

GridSearchCV on Task 2's best RF (CNN + traditional, 8 combinations x 5-fold = 40 fits) yielded  $n\_estimators=200$ ,  $min\_samples\_leaf=3$ ,  $max\_depth=None$ ,  $max\_features='sqrt'$  as optimal, improving macro-F1 by only +0.25 pp over the default (0.8628 to 0.8653).

## 4. Discussion

### 4.1 Why models and features differ across the two tasks

Following the three contrasts of Section 2.1, we explain each pattern in turn.

The CNN embedding gain is larger on Task 2 (+52 pp) than on Task 1 (+37 pp) because the baseline starts differ. Task 1 baselines already reach 54.7% from low-level statistics, leaving less room for CNN. Task 2 baselines are only 34.1%, so the CNN's semantic representation becomes the principal discriminator. Feature value is relative to task difficulty: harder tasks benefit more from better features.

On CNN features, model differences are small on Task 1 ( $< 3$  pp; LR > RF > SVM all within 3 pp) because the embedding makes classes nearly linearly separable, leaving non-linear models little to add. On Task 2 the gap widens to 9 pp (RF 86.3% > LR 84.4% > SVM 77.5%); RF's bagging variance reduction matters under the small (417) sample regime.

HOG works on Task 1 (43%) but collapses on Task 2 (21%) because it encodes gradient and contour. Task 1's animals (butterfly, spider, horse) differ in overall shape, but Task 2's birds share a similar body plan.

Feature concatenation harms LR and SVC on Task 2 because 219 dimensions

with only 417 samples overfit, a case of the curse of dimensionality (see Section 4.3). RF avoids this through implicit feature subset selection.

Adding HSV features helps linear models slightly but degrades RF on Task 2. This is a small-scale illustration of the same pattern: any additional feature block tends to help variance-controlled linear models but only adds noise to RF, whose own feature selection already keeps the strongest signals.

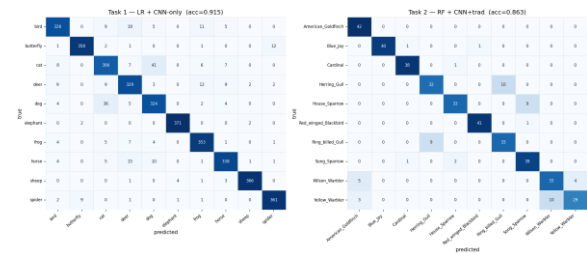
## 4.2 Which classes are most confused, and why

Task 1 (Figure 3, left): errors concentrate on the CIFAR-10 mammals. Cat-dog confusion alone accounts for about 24% of Task 1 errors ( $41 + 36 = 77/318$ ). Both classes are quadrupedal, furry, and similarly posed, so the CNN's global representation struggles with the subtle differences. In contrast, the four added classes (butterfly, elephant, sheep, spider) have distinctive shapes and reach  $F1 \geq 0.96$ .

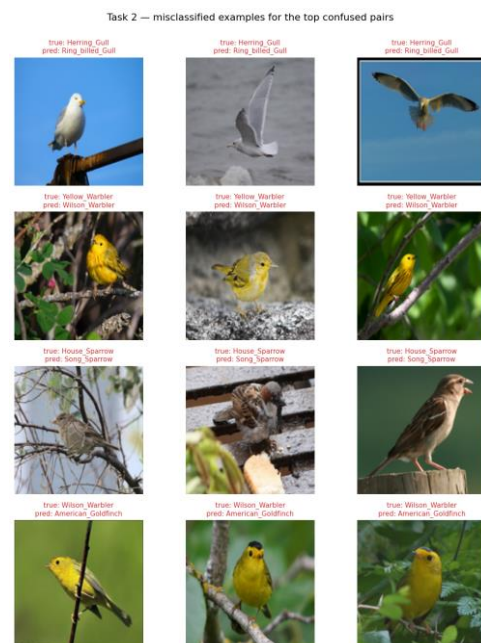
Task 2 (Figure 3, right): errors concentrate on four pairs, which are exactly the visually-confusing pairs we predicted from sample images in Section 1: Herring/Ring\_billed Gull, Wilson/Yellow Warbler, American\_Goldfinch/Yellow Warbler, and House/Song Sparrow. These four pairs account for 7 of the top-8 confusion entries on Task 2. Figure 4 shows that non-experts find these pairs hard to distinguish.

Per-class F1 confirms the pattern: classes with distinctive colors or shapes (Cardinal 0.96, Blue\_Jay 0.98, Red\_winged\_Blackbird 0.98) score much higher than classes that require local-detail discrimination (warblers and gulls,  $F1 =$

0.77-0.78). The residual errors on Task 2 reflect the data's inherent visual similarity, not classifier or feature deficiencies.



**Figure 3. Row-normalised confusion matrices for the best models. Off-diagonal mass concentrates on visually similar class pairs.**



**Figure 4. Three misclassified examples per pair for the top-4 confusion pairs on Task 2.**

## 4.3 Connection to course theoretical properties

The findings map cleanly onto ML theory.

Bias-variance trade-off: on Task 1 (3,750 samples), all classifiers fit complex patterns without serious overfitting, so model choice matters little and feature quality dominates. On Task 2 (417 samples), variance-sensitive linear models overfit while RF's bagging acts as implicit regularisation, explaining RF's consistent lead on Task 2.

Transfer-learning universality: ResNet50's +52 pp gain on CUB-200-2011 confirms that features learned from a large, diverse classification task (ImageNet) generalise even to fine-grained categories the network never saw during training.

Curse of dimensionality: adding 219-d manual features to Task 2's small training set degrades LR/SVC performance, a direct case of overfitting from parameter-to-sample imbalance. The remedy aligns with feature-selection principles: dimensionality should match sample size.

Occam's razor: once CNN features make classes approximately linearly separable, the simplest model (LR) wins, while complex models add variance without gaining discriminative signal.

#### **4.4 Coarse-to-fine progression: limitations and strengths**

Comparing Task 1 and Task 2 reveals the boundaries of three approaches.

Manual features handle coarse tasks (54.7%) but collapse on fine-grained ones (34.1%): low-level statistics cannot resolve differences that exist at the semantic level.

ImageNet-pretrained CNN transfer works on both (86%+) and gains more on fine-grained tasks, confirming that semantic features become relatively more valuable as task difficulty rises.

CNN global representations still ceiling on Task 2. The remaining 14% errors concentrate on visually near-identical pairs, indicating that average pooling discards localised detail (beak shape, wing spots) that fine-grained classification requires. Stacking features or tuning hyperparameters cannot break this ceiling.

Future work should preserve spatial information (part-based or attention-based models) rather than expand global features. Coarse-to-fine is not merely a harder version of the same problem; it demands different algorithmic capabilities, namely recognising overall class identity versus localising and comparing fine-grained details.

## **5. Conclusion**

This study evaluated two feature spaces (manual features and ResNet50 embedding) and three classifiers (LR, RBF-SVM, Random Forest) on a coarse-grained animal task and a fine-grained bird task.

Manual features reach 54.7% on Task 1 but only 34.1% on Task 2. Switching to ImageNet-pretrained ResNet50 embeddings raises both to 91.5% and 86.3%, with a larger absolute gain on the fine-grained task (+52 pp versus +37 pp). Error analysis shows that 7 of the 8 most-confused class pairs on Task 2 match our pre-modelling visual predictions, indicating that residual errors stem from inherent visual similarity rather than classifier or feature inadequacy.

Overall, feature quality affects performance far more than classifier choice, and this gap widens on fine-grained tasks. Going beyond the current ceiling on Task 2 requires architectures that preserve spatial information (part-based or attention mechanisms) rather than further work on global features.

## **6. References**

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